

Case Study: TechRetail Solutions Architecture

Members: Madhur Agrawal, Arsalan Malik, Dhiraj Paudel

Room - 3

Background



- TechRetail, a mid-sized retail company, wants to create a data pipeline to collect retail data from various sources, process it using advanced analytics, and visualize the results in a dashboard. The goal is to gain insights into sales trends and improve decision-making. The company wants to leverage Azure Databricks for data processing and Microsoft Fabric for data integration and visualization.

Objectives:



Data Ingestion



Data Processing

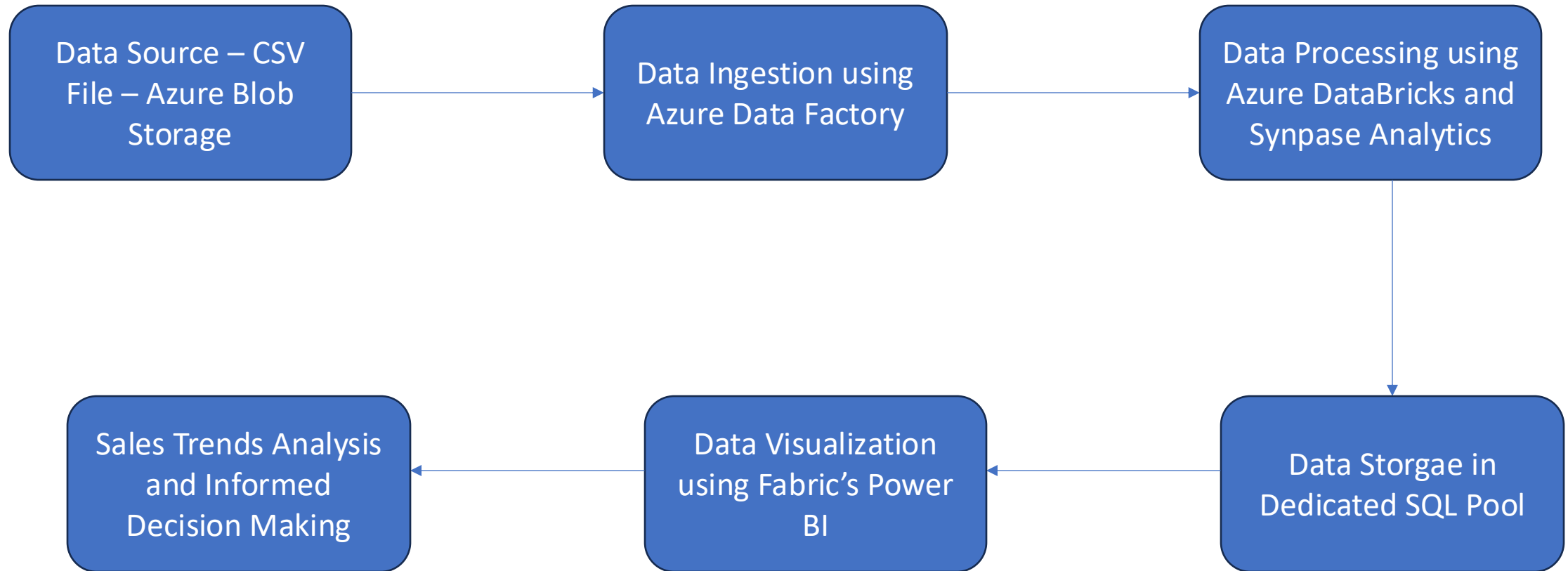


Data Storage



Data visualization

Solutions Architecture for TechRetail





Data Source – CSV Stored in Azure Blob Storage

The data is provided in the CSV format which is can be stored in the Azure Blob Storage. The Blob storage is a perfect choice to store the data which can be ingested with several services like Azure Data Factory.

The Blob Storage also provides a scalable and reliable storage solution.



Data Ingestion – Azure Data Factory (ADF)

- Using ADF to automate the data ingestion process from the CSV file.
- ADF can pull data from various sources, including Azure Blob Storage (if the CSV file is stored there) or on other on-premises locations if needed.
- Azure Blob Storage: This will be the staging area where CSV files are initially stored before ingestion. Blob Storage provides high scalability and security for storing raw data files.



Data Exploration and Cleaning

- Using Azure Synapse Analytics for quick data exploration and switch to Spark pools within Synapse to explore further data using Python.
- As data has been directly loaded into Azure Blob, it can be directly accessed within Azure Synapse Studio.
- Utilization of various SQL queries or Spark Notebooks that can handle missing values, duplicate values, data type corrections and other preprocessing task.



Data Processing – Azure Databricks & Azure Synapse Analytics

- Use Azure Databricks for advanced analytics and data processing. Databricks is ideal for handling large datasets, performing complex transformations, and running machine learning algorithms if required
- As an alternative to Databricks, Synapse offers integrated ETL capabilities with both Spark and SQL-based processing, seamlessly connecting with Azure Data Factory for transformation tasks.



Data Storage in the Dedicated SQL Pool of Synapse Analytics

- It serves as the storage layer for processed data, offering optimized analytics capabilities and scalable, multi-tier storage options. Azure Data Lake Storage is suitable for large datasets and complex queries.
- It can also store processed data in a SQL Pool for fast querying, ideal for analytical workloads that require quick SQL-based data access. It also integrates well with visualization tools like Power



Data Visualization using Microsoft Fabric's Power BI

- Microsoft's Fabric Platform provide Power BI solution which can be used for the visualization of the data which is analysed and processed by DataBricks and Synapse Analytics.
- It can directly connect to Azure Synapse which can be used to create the dashboards for the insights from the processed data.
- It can be used to monitor retail insights and trends in the data.
- Scheduled Refresh option is also available to update the visualization based on the updates with data



Analysis Performed over the Processed Data

- **Trend Analysis**

- How key things (sales, expenses, customer feedback) change over time
- Line charts to plot values over time and spot incline or decline pattern.

- **Top/Bottom Performers**

- Identification of top performing and underperforming products
- Basic table visual to rank entities based on sales and revenue followed by sorting to spot the highest and lowest values.



Analysis Performed over the Processed Data

- **Growth Analysis**

- Measurement of growth between two time periods (month – month , year – year)
- Simple calculations to show how sales have changed over specific period

- **Average Calculations**

- Average of a metric (average sales, average revenue , average ratings)
- Table to display the average of the metrics for further analysis



Analysis Performed over the Processed Data

- **Min / Max**

- Identification of smallest and largest values of the dataset
- Can be used to show max /mins like Maximum / Minimum Sales value.

- **Basic Proportion Analysis(Yes/No)**

- Customer feedback or cases can be resolved or unresolved , can be used to visualize what percentage of cases have been resolved and what are still open
- Pie chart to display the percentage of each outcome for quick and easy understanding of distribution.



Informed Decision Making

- The visualized data can be used for the informed decision making by the stakeholders and the business owners.
- The various market trends can be analysed using the interactive dashboards of Power BI which can help in making decisions to help improve sales and grow the business of TechRetail.